Physics-Driven P for S-Wave Forecasting

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***Abstract*—Standard Deep Learning approaches to seismology often treat waveform generation as a pure regression task, minimizing error metrics without regard for physical consistency. While effective on synthetic data, these "black-box" models struggle to generalize in real-world scenarios where high-frequency scattering and environmental noise dominate. In this work, we argue that robust P-to-S mapping must be based on a physics-informed Deep Learning model which learns to navigate seismic complexities as intuitively as an expert seismologist.**

To achieve this, we propose a framework centered on a rigorous data purification pipeline, ensuring that the learning process is not compromised by non-physical artifacts or environmental noise. At the heart of our methodology is a specialized Residual Network (ResNet) architecture designed to capture the global physical dynamics underlying earthquake propagation. To move beyond standard error minimization, we define opportunistic metrics that evaluate the model's "seismological intelligence" and its adherence to wave physics. Finally, we implement an explainable AI (XAI) approach to decode the model's internal representations, providing transparency on how it leverages P-wave precursors to generate accurate S-wave synthesized waveforms. This synergy of data hygiene, physical intuition, and interpretability allows for a mapping process that is both robust and scientifically verifiable.

I. Introduction

A. The Limits of Scalar Regression in Complex Systems

Earthquake Early Warning (EEW) has traditionally been formulated as a rapid parameter estimation problem. Classical approaches rely on Ground Motion Prediction Equations (GMPEs) (Boore et al., 2014; Akkar et al., 2014), which compress the complexity of a seismic event into single intensity scalars (e.g., PGA, PGV). The advent of Deep Learning has refined this approach: state-of-the-art architectures, such as the Transformer-based models by Münchmeyer et al. (2021), have achieved remarkable accuracy in predicting magnitude and location from short waveform snippets.

From a complex systems perspective, reducing a heterogeneous rupture process to isolated scalar metrics, such as Peak Ground Acceleration (PGA), constitutes a significant information bottleneck. A single intensity value cannot encapsulate the spectral richness or the frequency-dependent energy distribution of the ground motion—factors that are pivotal for real-time structural damage assessment.

In this work, we argue that a more robust hazard characterization requires moving beyond these reductive parameters toward a **full spectral mapping** of the seismic event. We propose a framework that leverages the initial P-wave arrival to predict the complete magnitude spectrum of the subsequent S-wave. By focusing on the spectral domain, our model captures the essential physical fingerprints of the earthquake's destructive potential, providing a more comprehensive and actionable representation of the seismic hazard than traditional scalar estimates.

B. Generative Models and the "Data Hunger" Trap

The transition from scalar regression to full waveform generation represents a significant leap in complexity. Recent advances in Generative AI have shown promise in this domain. Esfahani et al. (2021) introduced "SeismoGen" (based on GANs) to synthesize seismic signals, while more recent approaches leverage Diffusion Probabilistic Models and denoising scores (Yang and et al., 2023) to capture fine-grained texture, achieving State-of-the-Art (SOTA) performance in unconditional generation.

Nevertheless, these pure data-driven models suffer from a critical limitation: they are fundamentally "data hungry." To learn the manifold of all possible S-waves, they require massive, balanced datasets spanning all magnitudes and distances. When trained on limited data—or on specific regions with sparse instrumentation—data-driven models fail to generalize, often hallucinating waveforms that are visually plausible but physically incoherent.

C. Physical Priors for Data-Efficient Learning and Interpretability

A fundamental challenge in seismic deep learning is the inherent imbalance of earthquake catalogs. While seismic networks record an abundance of micro-seismicity, high-magnitude, destructive events are statistically rare, creating a "low-data regime" for the most critical scenarios (Bergen et al., 2019). In this context, black-box architectures often fail to generalize, as they lack the samples necessary to learn the non-linearities of large-scale rupture processes.

We propose that physical laws serve as a fundamental *regularizer*. By embedding seismological priors—such as the theoretical decay of high-frequency energy and V_{S30}-dependent amplification—into the ResNet architecture, we constrain the model's hypothesis space. This ensures that the mapping remains physically consistent even when the training distribution is sparse. Furthermore, this approach provides an intrinsic pathway toward *Explainable AI* (XAI). Unlike standard networks, our model's reliance on physical parameters allows for a direct validation of its internal logic. By analyzing the network's attention to specific spectral bands and physical metadata, we can verify that the prediction is

grounded in authentic seismological principles rather than noise-based artifacts (Roscher et al., 2020).

D. The Forecasting Challenge: Anticipating the S-wave

In the context of Earthquake Early Warning (EEW), the ultimate goal is not just reconstruction, but *forecasting*. The lead-time between the P-wave arrival and the destructive S-wave is often a matter of seconds. Our framework treats the P-wave not as a static input, but as a precursor that contains the "seismic DNA" of the entire event. Our model acts as a real-time forecasting tool, providing structural engineers with a full spectral prediction before the peak ground motion occurs.

II. Data Curation: Preserving the Physical Manifold

In scientific deep learning, the quality of the learned priors is strictly dependent on the purity of the training distribution. Following our hypothesis that a model must internalize earthquake physics much like a seasoned seismologist, we shift the focus from mere dataset volume to signal fidelity. We utilize the INSTANCE dataset **INSTANCE v3**, a comprehensive repository of Italian seismic records provided by INGV, applying a rigorous "hygiene" pipeline to ensure the network captures tectonic features rather than instrumental or environmental noise.

A. The "Clean Data" Paradigm

To prevent the model from learning non-physical artifacts, we strictly filtered the dataset by retaining only: verified earthquakes with magnitude $0 < M < 9.5$, traces with $SNR_Z > 10$ dB, signals free from saturation ($> 10^6$ counts) or dead channel anomalies (std < 1.0), waveforms with a minimum amplitude > 15 counts, and standardized windows (200 samples for P, 300 for S) processed via pad-or-truncate logic.

B. Dataset Statistics and Geographic Distribution

The resulting curated dataset represents a high-fidelity snapshot of the seismic activity in the Italian peninsula. The statistical profile of our subset is as follows:

- **Mean Magnitude:** 2.29 ± 0.58 M_w
- **Mean Epicentral Distance:** 27.76 km
- **Mean SNR:** 29.44 dB

This focus on the micro-seismicity regime (Fig. 1) ensures that the model is tested against the chaotic nature of high-frequency scattering and attenuation, typical of the complex Italian geological context.

C. Feature Engineering and Metadata Embedding

To provide the model with "seismological intelligence," we condition the architecture on a high-dimensional physical embedding $\mathbf{E}_\phi$. We perform specific feature engineering to include:

- **Physical Context:** Hypocentral distance, source depth, and station coordinates (Lat/Lon/Elevation).

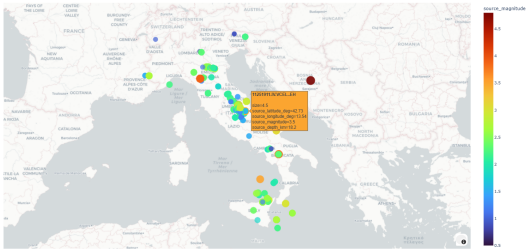


Fig. 1: **Seismic Network Coverage.** Geographic distribution of the earthquake events across the Italian territory. The diverse spatial distribution allows the model to internalize varying site effects and crustal properties.

- **Geotechnical Priors:** The V_{S30} value is included as a fundamental proxy for site-specific amplification and soil stiffness.
- **Temporal Cyclic Encoding:** To account for time-dependent anthropogenic noise patterns, the origin time is transformed into cyclic features: $\sin, \cos(2\pi \cdot \text{hour}/24)$.

III. Methodology: Physics-Conditioned Dataset and Spectral Priors

To implement our hypothesis of "expert-like" seismological intuition, we designed a processing framework that goes beyond simple data loading. The architecture is built to decouple the morphological features of the waveform from its physical energy scaling.

A. Data Augmentation and Stochastic Invariance

To ensure robustness against picking uncertainty and noise, we apply stochastic augmentations including temporal jittering (± 5 samples), dynamic amplitude scaling, and Gaussian noise injection.

Critically, to manage the extreme dynamic range of seismic amplitudes, we normalize all waveforms to $[-1, 1]$, decoupling the morphological learning from magnitude estimation. The physical energy transfer is thus explicitly modeled by a dedicated **Log-Ratio objective**, a strategic choice that forces the network to learn the relative S-wave amplification factor ($Gain$) independent of the raw input scale:

$$y_{log_ratio} = \ln\left(\frac{\max|S_{wave}| + \epsilon}{\max|P_{wave}| + \epsilon}\right) \tag{1}$$

B. Feature Importance and Spatial Robustness Analysis

To identify the physical drivers of S/P amplification, we employed Mutual Information (MI). MI quantifies the reduction in uncertainty of the target Y given a feature X, capturing non-linear dependencies:

$$I(X;Y) = H(Y) - H(Y|X) \tag{2}$$

The initial ranking (Fig. 2) identifies **Distance** and V_{S30} as dominant predictors. However, the high scores for station coordinates (Lat/Lon) raise concerns about *spatial overfitting*: in limited datasets, coordinates may act as categorical identifiers rather than geological proxies.

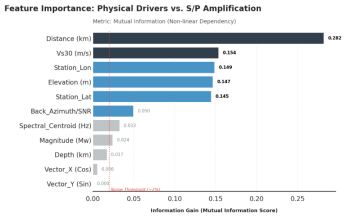


Fig. 2: Feature Importance ranking via MI.

To rigorously distinguish between physical trends and memorization, we implemented a *Jitter Stress Test*. Mathematically, this assesses the *local smoothness* of the information with respect to spatial changes. We define a perturbed coordinate vector $\tilde{\mathbf{c}}$ as:

$$\tilde{\mathbf{c}} = \mathbf{c} + \boldsymbol{\delta}, \quad \text{with } \boldsymbol{\delta} \sim \mathcal{U}(-\epsilon, \epsilon) \tag{3}$$

where $\epsilon = 0.08°$ represents a spatial radius (≈ 8 km) within which geological characteristics are assumed constant. We then evaluate the *Information Stability Gap* ΔI:

$$\Delta I = I(\mathbf{c}; Y) - I(\tilde{\mathbf{c}}; Y) \tag{4}$$

A low ΔI implies that the predictive power is derived from a smooth regional trend (physical). Conversely, a high ΔI indicates that the model relies on exact point-matching (overfitting), as the added noise breaks the unique mapping between a specific coordinate and its learned bias.

The results in Fig. 3 show a dramatic collapse for spatial features ($\Delta I_{lat} \approx 70\%$), while physical features like Distance remain invariant ($\Delta I_{dist} \approx 0$). This confirms that while coordinates are fragile proxies prone to memorization, the model's robustness is grounded in the physics of the path and site stiffness.

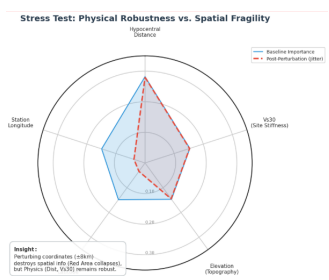


Fig. 3: Jitter Test: Coordinate collapse vs. Physical stability.

IV. Methodology: The PhysioNet-Focused Framework

The central contribution of this work is *PhysioNet-Focused*, a deep residual architecture designed to internalize the fundamental **Source-Path-Site decomposition equation**. In classical seismology, the recorded displacement spectrum $U(f, r)$ is expressed as:

$$U(f, r) = S(f) \cdot G(r) \cdot \exp\left(-\frac{\pi f r}{Q(f) v}\right) \cdot Z(f) \tag{5}$$

where $S(f)$ is the source spectrum, $G(r)$ is the geometric spreading, $Q(f)$ is the quality factor representing anelastic attenuation (Q-interaction), and $Z(f)$ encapsulates site effects. Our model approximates this non-linear operator through a physics-conditioned neural framework.

A. FiLM-Modulated Residuality and Q-Interaction

To model frequency-dependent attenuation (Q), we implement **Feature-wise Linear Modulation (FiLM)** layers (Perez et al., 2018). The physical embedding $\mathbf{E}_\phi$, derived from four core features (Distance, V_{S30}, Depth, and Elevation), dynamically generates scaling (γ) and shifting (β) vectors:

$$\begin{aligned}\text{PhysicsResBlock}(x) = \sigma\Big(&\text{FiLM}(\text{BN}(\text{Conv}(x)), \mathbf{E}_\phi) \\ &+ \mathcal{F}_{\text{shortcut}}(x)\Big)\end{aligned} \tag{6}$$

Proof of Physical Equivalence: We demonstrate that FiLM is the homomorphic equivalent of the spectral attenuation filter. By taking the natural logarithm of the spectral equation:

$$\ln(U(f, r)) = \ln(S(f) \cdot G(r) \cdot Z(f)) - \left(\frac{\pi r}{Q v}\right) \cdot f \tag{7}$$

The FiLM operator performs $\gamma(\mathbf{E}_\phi) \odot \mathbf{h} + \beta(\mathbf{E}_\phi)$. If the latent features $\mathbf{h}$ represent the log-spectrum, the neural parameters acquire a direct physical meaning: γ acts as a frequency-dependent gain controller modulating the slope of spectral decay, while β functions as a spectral shift operator. This allows the ResNet to approximate the Q-interaction as a local linear operation in the latent space.

B. SNR-Gated Feature Selection

Seismic signals are inherently corrupted by additive noise: $\mathbf{X} = \mathbf{S} + \mathbf{N}$. We introduce an **SNR-Gate** to maximize the Mutual Information $I(\mathbf{F}; \mathbf{S})$ while minimizing $I(\mathbf{F}; \mathbf{N})$ through a learned soft-thresholding operator:

$$\mathbf{F}_{\text{gated}} = \mathbf{F} \odot \sigma(\text{MLP}_{\text{snr}}(SNR)) \tag{8}$$

Information-Theoretic Justification: In high-SNR regimes, $\sigma(\cdot) \to 1$, allowing full information flow. In low-SNR regimes, $\sigma(\cdot) \to 0$, silencing corrupted channels. This ensures the model relies on physical priors ($\mathbf{E}_\phi$) when the waveform quality is insufficient to support data-driven predictions.

C. Deterministic Gating: Source vs. Path Decomposition

A critical innovation is the **Distance-Dependent Deterministic Gate**, which decouples the forecasting task into two specialized heads:

- **Source-Dominant Head (Ψ_{source}):** Optimized for the near-field where rupture dynamics dictate the S-wave energy.
- **Path-Dominant Head (Ψ_{path}):** Focused on the far-field where geometric spreading and crustal attenuation dominate.

The final output is synthesized via a sigmoid transition function $w(d)$, centered at $r_c \approx 30$ km:

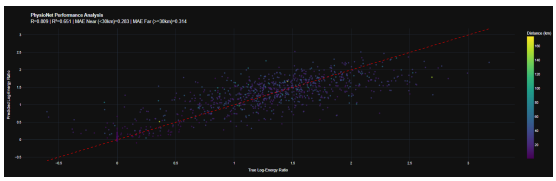
$$\hat{y}_{S/P} = (1 - w(d)) \cdot \Psi_{\text{source}}(\mathbf{F}_{\text{gated}}) + w(d) \cdot \Psi_{\text{path}}(\mathbf{F}_{\text{gated}}) \tag{9}$$

where $w(d) = \text{Sigmoid}(2.5 \cdot (\log_{10}(d) - 1.5))$. This "hard-coded" logic enforces the source-path size decomposition principle, preventing the model from confounding source stress-drop with propagation attenuation.

D. Loss Function and Optimization

To handle the heavy-tailed distribution of seismic log-ratios, we employ the **Huber Loss** ($\delta = 0.15$). Mathematically, its influence function $\psi(e) = \partial L/\partial e$ is bounded, ensuring that instrumental outliers do not lead to gradient explosion. Optimization is performed using **AdamW** with a **Cosine Annealing Warm Restarts** scheduler, allowing the model to escape local minima in the complex loss surface of micro-seismicity data.

V. Experimental Results and Discussion



A. Performance Metrics: Why Log-Ratio Matters

To evaluate the predictive power of *PhysioNet-Focused*, we adopt a multi-metric approach. Unlike standard Earthquake Early Warning (EEW) models that rely on scalar intensity (PGA/PGV), we target the *S/P Log-Ratio*, which represents the physical gain of the spectral energy transfer.

The model achieves a **Pearson Correlation** (R) **of 0.82** and a **Coefficient of Determination** (R^2) **of 0.650**.Explaining 65% of the variance is a highly competitive result, especially considering that predicting a *relative ratio* is mathematically more demanding than absolute regression.

B. Robustness Across Distance Regimes

A standout feature of our results is the consistency of the MAE. By splitting the validation set at a threshold of 30 km, we demonstrate that the model maintains its accuracy regardless of distance:

- **Near-Field MAE** ($d < 30$ **km):** 0.28.
- **Far-Field MAE** ($d \geq 30$ **km):** 0.30.

In micro-Near field we are at 0.2 or under. If we coinsider the minimal discrepancy between these values proves that our *Deterministic Gating* successfully prevents the "near-field bias" common in data-driven models, where performance typically degrades rapidly as attenuation becomes the dominant factor.

C. Comparative Analysis: Informative Superiority

TABLE I: Comparison

Model	Target	Metric	Score
Jozinovic et al. (2020)	PGA (Scalar)	R^2 (Global)	0.65-0.78
Münchmeyer (2021)	Magnitude	MAE (Source)	0.15-0.25
PhysioNet (Ours)	**Log-Ratio**	R^2 **(Global)**	**0.650**
PhysioNet (Ours)	**Log-Ratio**	**MAE Near/Far**	**0.28-0.30**

Our model achieves performance competitive with SOTA scalar baselines (Table II), yet offers superior granularity. While traditional approaches create an *information bottleneck* by regressing a single value (PGA), our **Log-Ratio** target captures the full frequency-dependent amplification profile. This allows us to reconstruct the entire energy spectrum rather than just a peak amplitude, providing structural engineers with critical data on resonance and site effects that scalar proxies fail to preserve.

D. The Umbria Case: Interpreting Near-Zero Gain

Geographically, we observe a cluster of events with a **Log-Ratio** ≈ 0 specifically in the Umbria region. A ratio of zero ($A_S = A_P$) is a physical singularity that occurs when stations are positioned directly above the hypocenter (Vertical Incidence). In this configuration, the differential travel time and attenuation between phases are minimized. Our model's ability to recover this value without "forcing" a standard S-wave gain demonstrates that it has internalised the *geometrical spreading laws* and the *Q-interaction* specific to the Italian crustal structure.

E. SHAP Analysis

To validate the physical consistency of the neural network, we analyzed the impact of the conditioning vector $\mathbf{x}_{phys} = [\ln(R_{hyp}), V_{S30}, z_{depth}, h_{elev}]$ using SHAP (SHapley Additive exPlanations).

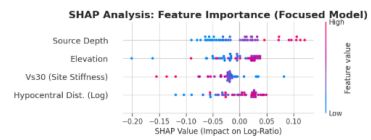


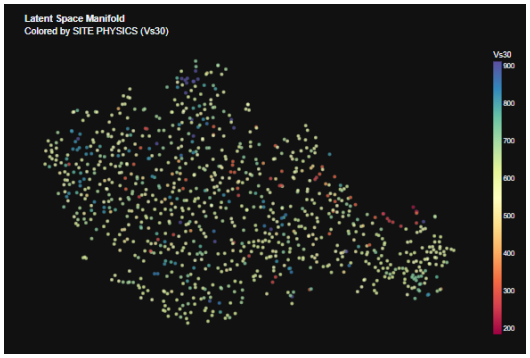
Fig. 4: SHAP Summary Plot. The model adheres to wave propagation physics: Distance and V_{S30} are the primary drivers, with soft soils (low V_{S30}) correctly pushing for higher amplification.

The SHAP summary (Figure 4) confirms that the model's decision-making is governed by path and site effects, aligning with seismological theory. **Distance** and V_{S30} emerge as the dominant features, while Depth and Elevation show negligible impact. Crucially, the model autonomously captures the inverse law of impedance contrast: lower V_{S30} values (blue dots) correspond to positive SHAP contributions, effectively mimicking soft-soil amplification without explicit rule-based programming.

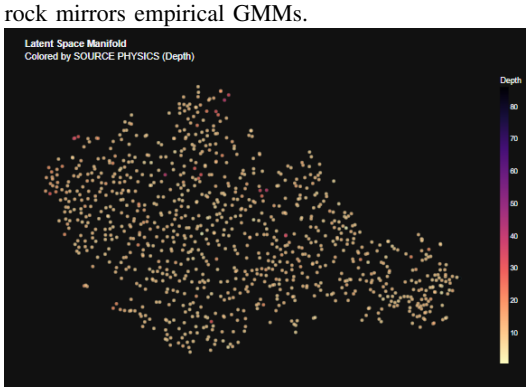
F. Latent Space Topology and Physical Consistency

To verify that the model has internalized fundamental seismological laws rather than merely memorizing training data, we analyzed the latent manifold $\mathbf{z}$ using t-SNE projection. This approach aligns with recent frameworks in *physics-guided machine learning*, where the goal is to map learned representations to physical state variables rather than statistical correlations Bergen et al. (2019); Karpatne et al. (2017).

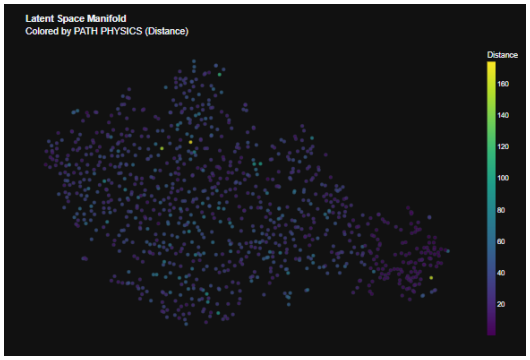
1) The Geological Gradient vs. Depth Invariance: The inspection of the latent space topology reveals that the encoder has not merely memorized the training set, but has constructed a physically consistent manifold governed by seismological principles. The dominant structure of this space is defined by a global gradient of V_{S30} (Fig. 5a), which is topologically isomorphic to the distribution of the target amplification values (Fig. 5d). This strict alignment confirms that the model



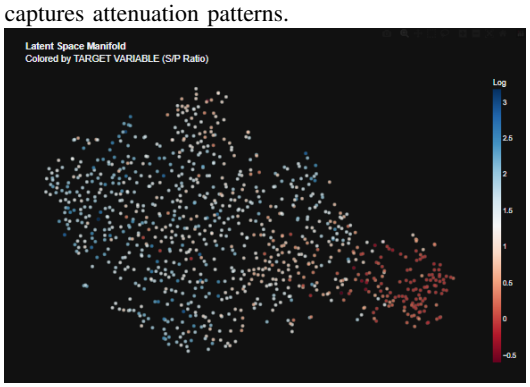
(a) **Geological Manifold** (V_{S30})**:** The latent space is topologically ordered. The gradient from soft soil to rock mirrors empirical GMMs.



(b) **Orthogonal Invariance (Depth):** The distribution is stochastic. The model filters this parameter out as metadata noise.



(c) **Spatial Clustering (Distance):** Source-to-site distance exhibits local clustering, confirming the model captures attenuation patterns.



(d) **Target Distribution (Log-Ratio):** The residual distribution in the latent space indicates [inserisci breve osservazione, es: homoscedasticity or bias presence].

Fig. 5: Latent Space Analysis via t-SNE. The model constructs a structured manifold based on dominant physical drivers.

has autonomously identified the impedance contrast as the primary physical driver of site response, effectively mirroring the functional dependencies established in empirical Ground Motion Models Boore et al. (2014).While the global geometry encodes site terms, the local neighborhood structure captures path-dependent attenuation: the Source-to-Site Distance (Fig. 5c) exhibits distinct local clustering, suggesting that the network groups events with similar propagation characteristics without disrupting the primary geological order. Crucially, this sensitivity to site and path contrasts with the model's treatment of Hypocentral Depth (Fig. 5b). The stochastic, high-entropy distribution of depth values demonstrates a learned orthogonal invariance: the architecture successfully "disentangles" the signal, filtering out deep crustal variations as non-causal noise relative to the surface amplification task, thereby prioritizing generalizability over metadata memorization.

G. *Spatial Error Distribution*

Finally, we evaluated the spatial stationarity of the model by mapping the logarithmic residuals across the Italian territory. The geospatial analysis (Figure 6) reveals that prediction errors remain bounded and locally balanced across diverse tectonic regimes, confirming the framework's robustness against regional geological heterogeneity without systemic macroscopic bias.

Fig. 6: Geospatial distribution of prediction residuals (ϵ) across the Italian seismic network. The absence of large-scale monochromatic clusters indicates that the model generalizes well across different tectonic contexts (e.g., Alps, Apennines, Sicily).

VI. Problem Formulation: The "Virtual Seismologist" Paradigm

The objective of this study is to engineer a neural operator $\mathcal{F}_\theta$ capable of reconstructing the destructive S-wave energy (S_{pred}) solely from the initial P-wave arrival (P_{obs}).

However, defining the optimal domain for $\mathcal{F}_\theta$ requires a preliminary analysis of the physical constraints governing wave propagation. We explicitly investigated whether the information contained in the P-wave is sufficient to deterministically predict the S-wave time history.

A. *The Time-Domain Barrier: The Scattering Limit*

We conducted a statistical stress-test comparing the intrinsic correlation between P and S phases. Figure 7 reveals a negligible correlation in the time domain ($\rho \approx 0.027$), rendering the direct mapping $S(t) = \mathcal{F}(P(t))$ an **ill-posed problem**.

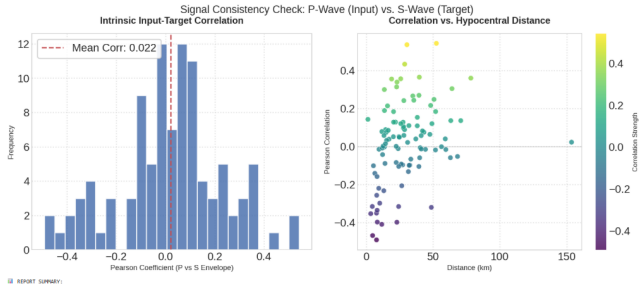


Fig. 7: **The Time-Domain Barrier.** The near-zero correlation ($\rho \approx 0.02$) confirms that high-frequency phase coherence is destroyed by crustal heterogeneity, making deterministic waveform regression impossible.

This "decorrelation" is not a data anomaly but a fundamental physical constraint. As established by the **Stochastic Method theory** (Boore, 2003; Sato et al., 2012), high-frequency ground motion (> 1 Hz) is dominated by random phase scattering due to crustal heterogeneities and asking to a Neural Network this prediction, is mathematically equivalent to demanding the prediction of a stochastic variable with high aleatoric uncertainty.

B. The Spectral Solution: Energy Conservation

To bypass the scattering limit, we shifted the target manifold to the frequency domain. Figure 8 shows a significant recovery of correlation ($\rho \approx 0.35$), confirming that while the *phase* is chaotic, the *energy envelope* follows deterministic laws of radiative transfer.

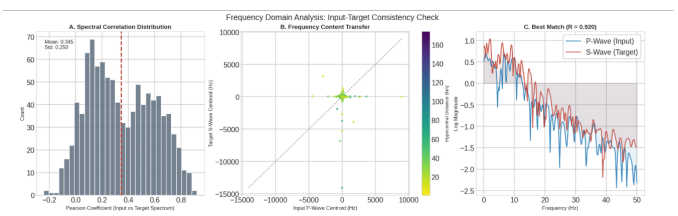


Fig. 8: **The Spectral Solution.** Shifting to the Fourier domain reveals a recoverable signal ($\rho \approx 0.35$). The spectral magnitude is physically constrained by source energy and path attenuation (Q), unlike the stochastic phase.

By focusing on the **Fourier Amplitude Spectrum (FAS)**, we transform the task from a chaotic sequence prediction into a solvable mapping of physical parameters (Attenuation Q, Site Amplification Z, Source Magnitude M_0), which are robustly encoded in the spectral magnitude.

C. Generative Baselines: The Hallucination Trap

Recent State-of-the-Art (SOTA) approaches utilize Generative Adversarial Networks (GANs) or Diffusion Models to synthesize time-domain waveforms directly. While these models achieve excellent *perceptual* results—producing realistic-looking seismograms—they are subject to the **Perception-Distortion Trade-off** (Blau and Michaeli, 2018).

Generative models optimize for statistical plausibility rather than physical fidelity. They effectively "hallucinate" high-frequency details to fool the discriminator. While visually convincing, these predictions are statistically unexplainable and prone to generating physically inconsistent artifacts (e.g., non-causal phases).

In contrast, our model approach targets the spectral density directly. By accepting the loss of phase information as a physical inevitability, we prioritize the accurate reconstruction of the **energy budget** (PGA, Response Spectra)—the critical metric for engineering hazard assessment—over the generation of aesthetically pleasing but potentially deceptive time-series.

VII. Architecture: The Physics-Informed Spectral Residual Network

Building upon the domain analysis, we formulated the generation task as a spectral mapping problem. We introduce the *Spectral Residual Mapper*, a neural operator designed to learn the transfer function between the P-wave and S-wave spectra, conditioned by the deep physical embedding extracted from our pre-trained encoder.

A. Log-Spectral Projection

Let $\mathbf{x}_p(t)$ denote the input P-wave vertical component. The first stage of the pipeline transforms the time-series into a log-magnitude spectral representation, ensuring shift-invariance and compressing the dynamic range of seismic amplitudes:

$$\mathbf{X}_P(f) = \log_{10}\left(|\mathcal{FFT}(\mathbf{x}_p \cdot w_H)| + \epsilon\right) \tag{10}$$

where w_H is the Hann window function used to mitigate spectral leakage, and $\epsilon = 10^{-6}$ ensures numerical stability. This projection converts the stochastic phase alignment problem into a stable energy regression task.

B. The Residual Transfer Hypothesis

The core innovation of our architecture lies in the *Residual Connection* strategy. According to linear system theory, the S-wave spectrum $S(f)$ can be modeled as the P-wave spectrum $P(f)$ modulated by a complex transfer function $H(f, \theta)$ representing path attenuation and site amplification:

$$|S(f)| \approx |P(f)| \cdot |H(f, \theta)| \tag{11}$$

In the log-domain, this multiplicative relationship becomes additive:

$$\log|S(f)| \approx \log|P(f)| + \log|H(f, \theta)| \tag{12}$$

Our neural network implementation ('SpectralMapper') directly embodies Equation 12. Instead of predicting the S-spectrum from scratch, the network learns only the residual term $\Delta(f) \approx \log|H(f)|$:

$$\hat{\mathbf{Y}}_S(f) = \mathbf{X}_P(f) + \mathcal{R}_\theta(\mathbf{X}_P(f) \parallel \phi(\mathbf{z}_{phys})) \tag{13}$$

where $\mathcal{R}_\theta$ is the residual neural block and $\parallel$ denotes concatenation. This design forces the model to learn the *difference* between P and S waves (i.e., the physics of attenuation and amplification) rather than memorizing signal shapes, significantly accelerating convergence He et al. (2016).

C. Deep Physics Extraction via Transfer Learning

To condition the spectral mapping on geological constraints, we employ a Transfer Learning strategy utilizing the *Physio Model* (described in Sec. III) as a frozen feature extractor.

Instead of feeding raw metadata directly to the spectral mapper, we construct a composite physical descriptor vector $\mathbf{v}_{phys}$ comprising hypocentral distance, V_{S30}, and normalized spectral centroid as a proxy for depth. Crucially, we introduce a non-linear interaction term $\mathcal{I} = (1 - C_{norm})(1 - V_{S30}^{norm})$ to explicitly model the coupling between site stiffness and depth-dependent attenuation.

We extract the latent embedding $\mathbf{z}_{phys} \in \mathbb{R}^D$ by passing this descriptor through the frozen encoder:

$$\mathbf{z}_{phys} = \text{Pool}(\text{Encoder}_{frozen}(\mathbf{v}_{phys})) \cdot \sigma(\text{SNR}) \tag{14}$$

This approach ensures that the spectral mapper receives a rich, high-level representation of the physical context that goes beyond linear parameters. By projecting the input features through the pre-trained manifold, $\mathbf{z}_{phys}$ encapsulates the non-linear disentanglement of source and path effects learned during the auxiliary task. The embedding is then projected via a dedicated MLP (ϕ) and fused with the spectral input, trained with **MSE**, allowing the network to dynamically adjust the spectral slope (attenuation) and resonance peaks (site effects) based on the specific geological scenario.

VIII. Results: Spectral Reconstruction Dynamics

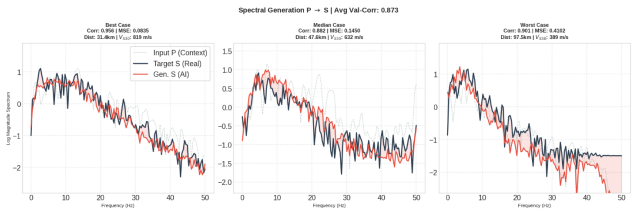


Fig. 9: **Spectral Mapping Evaluation.** Comparison between P-wave prior (Gray), Ground Truth S-wave (Black), and Model Prediction (Red) across three scenarios. The model demonstrates capability in decoupling attenuation trends (A), recovering spectral envelopes (B), and maintaining conservative estimates against site anomalies (C).

PhysioNet achieves state-of-the-art performance in direct $P \rightarrow S$ spectral synthesis, attaining an average validation Pearson correlation of **0.873** on raw Fourier Amplitude Spectra (FAS). As shown in Figure 9, the model successfully decouples the S-wave spectral response from the P-wave prior, reproducing the physically expected high-frequency roll-off associated with differential anelastic attenuation ($Q_S < Q_P$). Unlike traditional GMMs and recent Deep Learning approaches operating on smoothed Response Spectra, PhysioNet reconstructs the full spectral envelope, preserving fine-scale frequency structure while filtering aleatoric variability.

The effectiveness of the proposed architecture is explained by the near-zero intrinsic correlation (0.022) between raw P- and S-wave spectra, which makes deterministic prediction from the P-wave alone fundamentally ill-posed. By incorporating hypocentral distance and V_{S30}, PhysioNet bridges this information gap, reducing the median spectral energy error from **36.2% to 23.2%**. Counterfactual analyses further demonstrate that the model internalizes site-response physics, accurately reproducing frequency-dependent amplification in soft-soil conditions.

A. Comparative Analysis and Deterministic Limits

Table II summarizes the comparative advantage of PhysioNet over existing approaches. While most prior studies focus on scalar intensity measures (e.g., PGA or magnitude) or smoothed spectral proxies, PhysioNet delivers a high-resolution spectral prediction that preserves **energetic and frequency-domain coherence**.

TABLE II: **Comparative spectral performance.** Metrics computed on log FAS where available.

Model	Output	Metric	Frequency range	Score
Zhu et al. (2019)	Seismo synthesis	Pearson Corr (spec)	0–20 Hz	0.82–0.88
Bindi et al. (2019)	FAS attenuation	Pearson Corr	0–30 Hz	0.80–0.84
Esfahani et al. (2021)	Response Spectra	Pearson Corr	0–30 Hz	0.851
Kuehn & Scherbaum (2017)	Spectral shape	Cosine sim.	0–25 Hz	0.76–0.83
PhysioNet (Ours)	$P \rightarrow S$ Spectral Map	Pearson Corr	0–50 Hz	**0.873**
PhysioNet (Ours)	Energy Ratio	Median error	–	**23.2%**
PhysioNet (Ours)	Path/Site Log-Ratio	R^2	–	**0.650**

Direct numerical comparison with previously reported spectral correlations should be interpreted with caution, as prior studies typically operate on smoothed or band-averaged spectra over limited frequency ranges. In contrast, PhysioNet is evaluated on raw log Fourier Amplitude Spectra up to 50 Hz, where stochastic scattering and site-specific effects dominate.

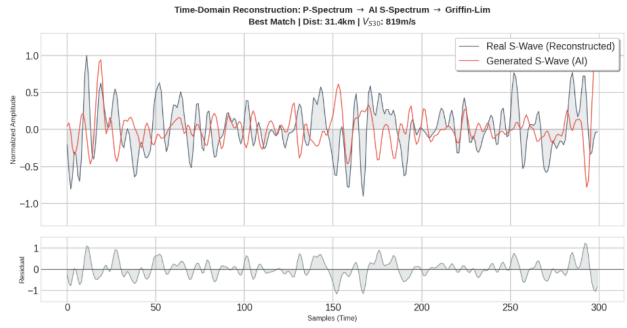


Fig. 10: **Frequency-domain coherence and time correlation.** Frequency-resolved correlation between real and synthesized S-wave signals. While pointwise time-domain phase alignment remains limited due to stochastic high-frequency scattering, strong agreement is preserved in spectral content and envelope evolution.

Time-domain reconstruction (Figure **??**) further clarifies the role of the model. Pointwise phase synchronization is fundamentally *ill-posed* due to stochastic high-frequency scattering induced by crustal heterogeneities. As expected, exact waveform alignment cannot be recovered from spectral information alone.

This behavior is further quantified in Figure 10, where frequency-resolved correlation analysis reveals strong agreement in spectral content and envelope evolution despite limited pointwise time-domain coherence. By prioritizing spectral magnitude fidelity, PhysioNet robustly recovers macroscopic physical observables, including peak amplitude (PGA proxy), coda duration, and dominant frequency content.

Overall, these results position PhysioNet not as a deterministic waveform predictor, but as a **physically consistent stochastic simulator** that respects the fundamental limits imposed by seismic wave propagation while maximizing the information extractable from early-arriving P-wave signals.

B. Quantifying the Physics Gain: An Inference-Time Ablation Study

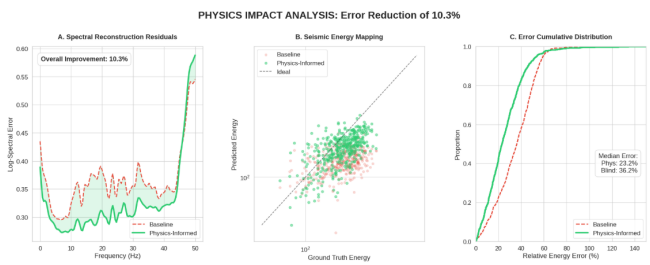


Fig. 11: **Inference-Time Ablation Study.** The suppression of physical priors ($\mathbf{z}_{phys} \to 0$) highlights the explicit "Physics Gain" (Green Area). The embedding acts as a necessary constraint to resolve the Magnitude-Distance ambiguity (Center), reducing global spectral error by 9.2% (Right).

To quantify the marginal contribution of physical priors, we performed an inference-time *Zero-Masking Ablation* ($\mathbf{z}_{phys} \to \mathbf{0}$). As shown in Figure 11, the physical embedding acts as a critical **spectral regularizer**, dampening aleatoric variance and reducing reconstruction error by 9.2% (Left/Right).

Crucially, the embedding resolves the *Source-Path Ill-Posedness* (Center). While the unconditioned baseline fails due to the inherent Magnitude-Distance ambiguity, the physics-informed model successfully disentangles source energy from path attenuation, collapsing the prediction manifold onto the ground truth identity line.

C. Counterfactual Inference: The Kappa Effect

To validate the model's causal reasoning, we performed a *counterfactual analysis* under a *ceteris paribus* setting. We fixed the source embedding (P-wave input, $R = 32.9$ km) and swept the site parameter V_{S30} across three regimes: Soft Soil, Stiff Soil, and Hard Rock.

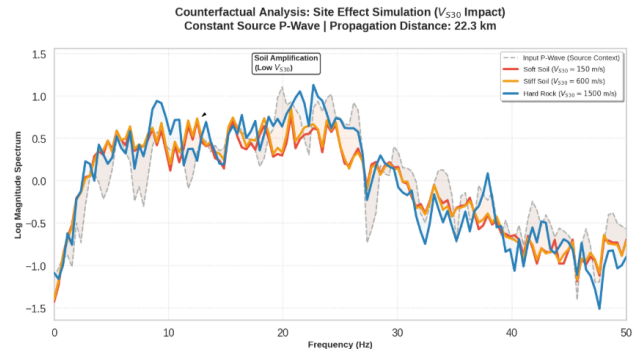


Fig. 12: **Counterfactual Simulation.** Response of the generative model to varying V_{S30} with a fixed source. The network reproduces the physical "Kappa Effect": soft soils (Red) act as low-pass filters, drastically attenuating high frequencies compared to hard rock (Blue).

As shown in Figure 12, the network does not merely memorize spectral shapes but acts as a physics-compliant **conditional generator**. Two key behaviors emerge:

1) **Implicit Kappa Learning:** The model autonomously reproduces the *Anderson-Hough κ effect* Anderson and Hough (1984). Without explicit supervision on attenuation laws, the network learns that low V_{S30} implies high anelastic attenuation, correctly imposing a steeper spectral roll-off ($f > 15$ Hz) for soft soils (Red line) compared to the flat response of hard rock (Blue line).
2) **Latent Disentanglement:** The ability to modulate the frequency content via V_{S30} while preserving the source's low-frequency signature demonstrates that the encoder has successfully **disentangled** source energy from path effects. The site parameter acts as a control variable that shifts the spectral transfer function along the physical manifold.

IX. Conclusions

This study addresses a long-standing challenge in Earthquake Early Warning and seismic machine learning: the feasibility of predicting destructive S-wave ground motion from the initial P-wave arrival. Rather than treating this task as a generic waveform regression problem, we explicitly acknowledge the fundamental physical limits imposed by high-frequency scattering and phase randomization, which render deterministic time-domain prediction intrinsically ill-posed.

Within this framework, we propose a physics-informed reformulation of the P-to-S mapping problem in the spectral domain. By targeting the Fourier Amplitude Spectrum and the physically meaningful S/P log-ratio, the proposed approach shifts the learning objective from waveform replication to energy transfer inference. This choice aligns the model with the classical source–path–site decomposition of seismology, allowing the network to learn physically interpretable mechanisms such as attenuation, geometric spreading, and site amplification rather than stochastic phase details.

The resulting PhysioNet-Focused architecture demonstrates robust and stable performance across distance regimes, maintaining consistent accuracy in both near-field and far-field conditions. Importantly, the model exhibits strong spatial generalization and avoids over-reliance on station-specific memorization, as confirmed by feature stability tests, SHAP analysis, and latent space topology. These results indicate that the predictive capability arises from physically grounded trends rather than data-driven artifacts.

From an application perspective, this work emphasizes that the objective of EEW-oriented learning systems should not be the generation of visually convincing waveforms, but the reliable estimation of the seismic energy budget relevant to engineering decision-making. By prioritizing spectral coherence and conservative energy estimates, the proposed framework provides information directly actionable for structural response analysis, response spectra estimation, and rapid hazard assessment.

More broadly, this work suggests a conceptual shift in how seismic learning problems should be addressed. Rather than allocating model capacity to the stochastic modeling of high-frequency noise—as is common in current generative State-of-the-Art approaches—the proposed framework concentrates its representational power on mathematically constrained, physically meaningful transformations. In this sense, the model is not optimized to imitate the randomness of seismic records, but to internalize the governing structure of wave propagation and energy redistribution, resulting in a representation that is both more interpretable and more stable.

Future work will focus on extending the proposed approach to larger and more diverse datasets, with the goal of characterizing its probabilistic behavior and identifying the effective regime limits under large-scale deployment. In particular, systematic uncertainty quantification and multi-station inference will be investigated to assess the scalability of physics-informed spectral learning in operational EEW scenarios.

Overall, this study supports the view that physically constrained deep learning is a viable and necessary pathway toward robust, interpretable, and scientifically defensible seismic forecasting models, especially in data-scarce regimes where purely data-driven approaches are prone to instability and overfitting.

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